

Math Exploration 1

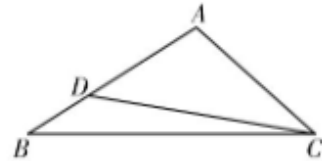
2. As shown in the figure, $\triangle ABC \sim \triangle ACD$, $AB = 10\text{cm}$, and $AC = 8\text{cm}$. The length of $\overline{AD}$ is _____.

A. 6.4cm

B. 6cm

C. 2cm

D. 4cm

**Math Exploration 1**

2. As shown in the figure, $\triangle ABC \sim \triangle ACD$, $AB = 10\text{cm}$, and $AC = 8\text{cm}$. The length of $\overline{AD}$ is _____.

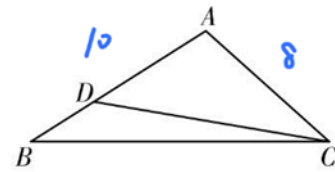
A. 6.4cm

B. 6cm

C. 2cm

D. 4cm

$$\frac{\triangle ABC}{\triangle ACD} = \frac{BC}{CD} = \frac{AC}{AD} = \frac{AB}{AC}$$



$$\frac{8}{AD} = \frac{10}{8}$$

$$AD = 6.4$$



Math Exploration 2

2. ~~SAT~~ Among the following, the number of conditions that can be used to deduce that $\triangle ABC$ is similar to $\triangle A'B'C'$ is _____.

① the two angles of $\triangle ABC$ are, respectively, 58° and 70° ; the two angles of $\triangle A'B'C'$ are, respectively, 58° and 52° .

② the two side lengths of $\triangle ABC$ are, respectively, 4cm and $\frac{3}{2}\text{cm}$ and their included angle is 40° ; the two side lengths of $\triangle A'B'C'$ are $\frac{4}{3}\text{cm}$ and $\frac{1}{2}\text{cm}$ respectively, and their included angle is 40°

③ the side lengths of $\triangle ABC$ are 5cm , 6cm , and 8cm , respectively; the side lengths of $\triangle A'B'C'$ are $\frac{5}{2}\text{cm}$, 3cm , and 4cm , respectively.

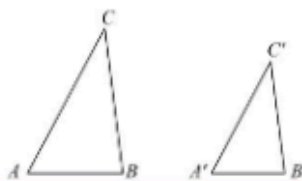
④ in $\triangle ABC$, $m\angle C = 90^\circ$, $AC = 3$, and $BC = 4$; in $\triangle A'B'C'$, $m\angle C' = 90^\circ$, $A'C' = 6$, and $B'C' = 8$.

A. 1

B. 2

C. 3

D. 4



Math Exploration 2

2. ~~SAT~~ Among the following, the number of conditions that can be used to deduce that $\triangle ABC$ is similar to $\triangle A'B'C'$ is _____.

① the two angles of $\triangle ABC$ are, respectively, 58° and 70° ; the two angles of $\triangle A'B'C'$ are, respectively, 58° and 52° . $180^\circ - 58^\circ - 70^\circ = 52^\circ$ AA

② the two side lengths of $\triangle ABC$ are, respectively, 4cm and $\frac{3}{2}\text{cm}$ and their included angle is 40° ; the two side lengths of $\triangle A'B'C'$ are $\frac{4}{3}\text{cm}$ and $\frac{1}{2}\text{cm}$ respectively, and their included angle is 40° . SAS $\frac{3}{2} \div 3 = \frac{1}{2}$

③ the side lengths of $\triangle ABC$ are 5cm , 6cm , and 8cm , respectively; the side lengths of $\triangle A'B'C'$ are $\frac{5}{2}\text{cm}$, 3cm , and 4cm , respectively. SSS

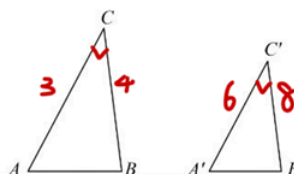
④ in $\triangle ABC$, $m\angle C = 90^\circ$, $AC = 3$, and $BC = 4$; in $\triangle A'B'C'$, $m\angle C' = 90^\circ$, $A'C' = 6$, and $B'C' = 8$. SAS ratio = 1:2

A. 1

B. 2

C. 3

D. 4



Math Exploration 3

2. In rectangle $ABCD$, $AB = 5$ and $BC = 3$. Points F and G are on $\overline{CD}$ so that $DF = 1$ and $GC = 2$. Lines $\overline{AF}$ and $\overline{BG}$ intersect at E . Find the area of $\triangle AEB$. (2003 AMC 10B Problem, Question #20)

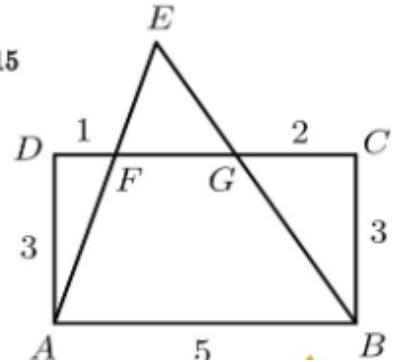
A. 10

B. $\frac{21}{2}$

C. 12

D. $\frac{25}{2}$

E. 15



2. In rectangle $ABCD$, $AB = 5$ and $BC = 3$. Points F and G are on $\overline{CD}$ so that $DF = 1$ and $GC = 2$. Lines $\overline{AF}$ and $\overline{BG}$ intersect at E . Find the area of $\triangle AEB$. (2003 AMC 10B Problem, Question #20)

A. 10

B. $\frac{21}{2}$

C. 12

D. $\frac{25}{2}$

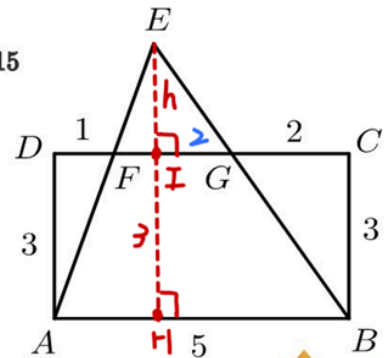
E. 15

$$\overline{EH} \perp \overline{FG}, \quad HI = 3, \quad EI = h$$

$$\overline{FG} \parallel \overline{AB}, \quad \triangle ABE \sim \triangle FGE$$

$$\frac{h}{h+3} = \frac{2}{5}, \quad h = 2$$

$$A = \frac{5 \times 5}{2} = \frac{25}{2}$$



Math Exploration 3

3. In $\triangle ABC$, $AB = AC = 28$ and $BC = 20$. Points D , E , and F are on sides $\overline{AB}$, $\overline{BC}$, and $\overline{AC}$, respectively, such that $\overline{DE}$ and $\overline{EF}$ are parallel to $\overline{AC}$ and $\overline{AB}$, respectively. What is the perimeter of parallelogram $ADEF$? (2013 AMC 12A Problem, Question #9)

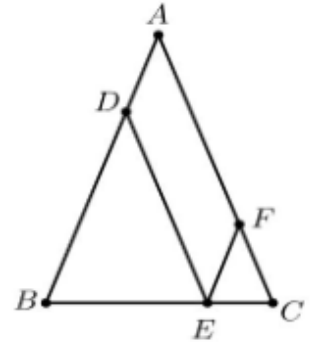
A. 48

B. 52

C. 56

D. 60

E. 72



$\Rightarrow$ iso $\triangle$

Math Exploration 3

3. In $\triangle ABC$, $AB = AC = 28$ and $BC = 20$. Points D , E , and F are on sides $\overline{AB}$, $\overline{BC}$, and $\overline{AC}$, respectively, such that $\overline{DE}$ and $\overline{EF}$ are parallel to $\overline{AC}$ and $\overline{AB}$, respectively. What is the perimeter of parallelogram $ADEF$? (2013 AMC 12A Problem, Question #9)

A. 48

B. 52

C. 56

D. 60

E. 72

$$\overline{DE} \parallel \overline{AC}, \overline{EF} \parallel \overline{AB}$$

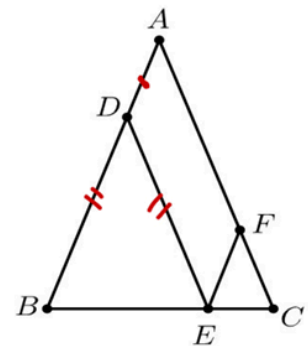
$$\triangle ABC \sim \triangle DBE \sim \triangle EFC \Rightarrow \text{iso } \triangle$$

$\downarrow$

$$DE = BD$$

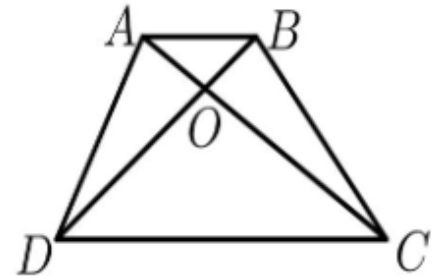
$$AB = AD + BD = 28$$

$$P = 28 \times 2 = 56$$



Math Exploration 4

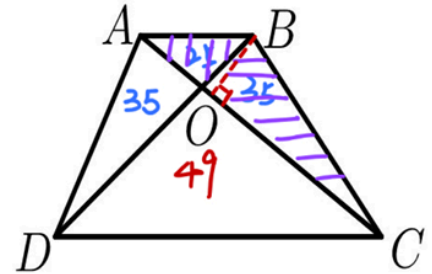
2. As shown in the figure, trapezoid $ABCD$ is divided into 4 smaller triangles by its diagonals. Given that the areas of $\triangle AOB$ and $\triangle BOC$ are 25cm^2 and 35cm^2 , respectively, then the area of the trapezoid is _____ cm^2 .



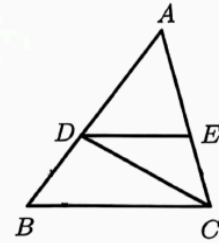
Math Exploration 4

2. As shown in the figure, trapezoid $ABCD$ is divided into 4 smaller triangles by its diagonals. Given that the areas of $\triangle AOB$ and $\triangle BOC$ are 25cm^2 and 35cm^2 , respectively, then the area of the trapezoid is 149 cm^2 .

$$\begin{aligned} \triangle AOB \sim \triangle COD &\Rightarrow \text{8-model} \\ \frac{AO}{CO} &= \frac{5}{7} & \frac{A_{\triangle AOB}}{A_{\triangle COD}} &= \frac{25}{49} \\ \frac{A_{\triangle AOB}}{A_{\triangle BOC}} &= \frac{\frac{1}{2} \cdot AO \cdot \text{height}}{\frac{1}{2} \cdot CO \cdot \text{height}} = \frac{25}{35} \end{aligned}$$



- 3 As shown in the figure, in $\triangle ABC$, point D and E are on $\overline{AB}$ and $\overline{AC}$. If $\overline{DE} \parallel \overline{BC}$, and $m\angle ACD = m\angle B$, then there are _____ sets of similar triangles in the figure.



Answer

4

Solution

Since point D and point E are on $\overline{AB}$ and $\overline{AC}$ respectively, and $\overline{DE} \parallel \overline{BC}$,
 $\triangle ADE \sim \triangle ABC$.

Since $m\angle ACD = m\angle B$ and $m\angle A = m\angle A$,

we have $\triangle ACD \sim \triangle ABC$,

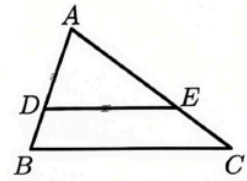
so $\triangle ADE \sim \triangle ACD$.

Since $m\angle B = m\angle DCE$ and $m\angle BCD = m\angle EDC$,

$\triangle DCE \sim \triangle CBD$.

Therefore, there are 4 sets.

- 5 As shown in the figure, in $\triangle ABC$, point D and point E are on $\overline{AB}$ and $\overline{AC}$, $\overline{DE} \parallel \overline{BC}$. If $AD = 2$, $AB = 3$, and $DE = 4$, then $BC = \underline{\hspace{2cm}}$.



Answer

6

Solution

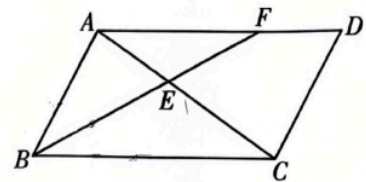
Since $\overline{DE} \parallel \overline{BC}$,
 $\triangle ADE \sim \triangle ABC$.

$$\text{Then } \frac{DE}{BC} = \frac{AD}{AB},$$

$$\text{so } \frac{4}{BC} = \frac{2}{3}.$$

Therefore $BC = 6$.

- 9 In parallelogram $ABCD$, the angle bisector of $\angle ABC$ intersects the diagonal $\overline{AC}$ at point E and intersects $\overline{AD}$ at point F . If $\frac{AB}{BC} = \frac{3}{5}$, then the value of $\frac{EF}{BF}$ is _____.



$$\frac{3}{8}$$

Solution

Since quadrilateral $ABCD$ is a parallelogram,

we have $\overline{AD} \parallel \overline{BC}$,

thus $m\angle AFB = m\angle EBC$.

Since $\overline{BF}$ is the angle bisector of $\angle ABC$,

we have $m\angle EBC = m\angle ABE = m\angle AFB$,

then $AB = AF$,

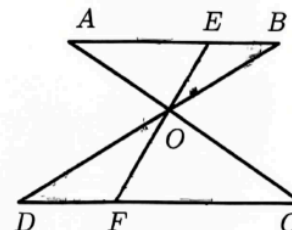
thus $\frac{AB}{BC} = \frac{AF}{BC} = \frac{3}{5}$.

Since $\overline{AD} \parallel \overline{BC}$,

we have $\triangle AFE \sim \triangle CBE$,

therefore $\frac{AF}{BC} = \frac{EF}{BE} = \frac{3}{5}$, $\frac{EF}{BF} = \frac{3}{8}$;

- 10 In the figure, $\overline{AB} \parallel \overline{CD}$, $\overline{AC}$ and $\overline{BD}$ intersect at point O . The line passing through point O intersects $\overline{AB}$ and $\overline{CD}$ at points E and F , respectively. The conclusion that may not necessarily be true is _____.



- A. $\frac{OA}{OC} = \frac{AB}{CD}$ B. $\frac{OA}{OD} = \frac{OB}{OC}$ C. $\frac{CD}{DF} = \frac{AB}{BE}$ D. $\frac{OE}{OF} = \frac{AB}{CD}$

B

Solution

Option A: Since $\overline{AB}$ is parallel to $\overline{CD}$, we have $\triangle ABO \sim \triangle CDO$. Therefore, $\frac{OA}{OC} = \frac{AB}{CD}$. So option A is correct.

Option B: Since $\triangle ABO \sim \triangle CDO$, we have $\frac{OA}{OC} = \frac{OB}{OD}$. However, we cannot conclude that $\frac{OA}{OD} = \frac{OB}{OC}$. Therefore, option B is incorrect.

Option C: Since $\triangle ABO \sim \triangle CDO$, we have $\frac{AB}{CD} = \frac{OB}{OD}$. Also, since $\overline{AB}$ is parallel to $\overline{CD}$, we have $\triangle OBE \sim \triangle ODF$. Therefore, $\frac{BE}{DF} = \frac{OB}{OD}$. Hence, $\frac{AB}{CD} = \frac{BE}{DF}$, which implies $\frac{CD}{DF} = \frac{AB}{BE}$. So option C is correct.

Option D: Since $\triangle OBE \sim \triangle ODF$, we have $\frac{OE}{OF} = \frac{OB}{OD}$. Also, since $\triangle ABO \sim \triangle CDO$, we have $\frac{AB}{CD} = \frac{OB}{OD}$. Therefore, $\frac{OE}{OF} = \frac{AB}{CD}$. Hence, option D is correct.
Therefore, the correct option is B.

Math Exploration 1

As shown in the figure, $\overline{AD} \parallel \overline{EF} \parallel \overline{BC}$, then

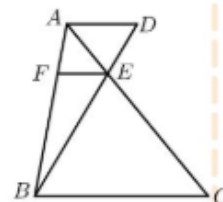
(1) $\frac{1}{AD} + \frac{1}{BC} = ?$; (2) if $AD = 3$, $BC = 5$, $AB = 12$, then calculate the length of $\overline{AF}$.

A. (1) $\frac{1}{EF}$ (2) $\frac{3}{8}$

B. (1) $\frac{1}{AF}$ (2) $\frac{9}{8}$

C. (1) $\frac{1}{AF}$ (2) $\frac{9}{2}$

D. (1) $\frac{1}{EF}$ (2) $\frac{9}{2}$



Math Exploration 1

As shown in the figure, $\overline{AD} \parallel \overline{EF} \parallel \overline{BC}$, then

(1) $\frac{1}{AD} + \frac{1}{BC} = ?$; (2) if $AD = 3$, $BC = 5$, $AB = 12$, then calculate the length of $\overline{AF}$.

A. (1) $\frac{1}{EF}$ (2) $\frac{3}{8}$

~~B. (1) $\frac{1}{AF}$ (2) $\frac{9}{8}$~~

~~C. (1) $\frac{1}{AF}$ (2) $\frac{9}{2}$~~

~~D. (1) $\frac{1}{EF}$ (2) $\frac{9}{2}$~~

Assume $AD = a$, $BC = b$

$$\frac{1}{AD} + \frac{1}{BC} = \frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab}$$

$$\triangle ADE \sim \triangle CBE \quad \frac{AD}{BC} = \frac{a}{b} = \frac{DE}{BE} = \frac{AE}{CE}$$



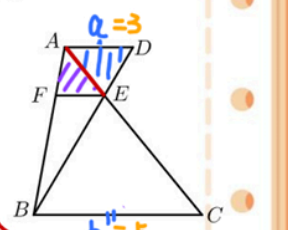
$$\triangle AFE \sim \triangle ABC \quad \frac{EF}{BC} = \frac{AE}{AC}$$

$$\frac{AD}{BC} = \frac{3}{5} = \frac{AE}{CE}, \quad \frac{AE}{AC} = \frac{3}{8}$$

$$\frac{EF}{b} = \frac{a}{a+b}, \quad EF = \frac{ab}{a+b} = a+b$$

$$\frac{1}{AD} + \frac{1}{BC} = \frac{1}{EF} = \frac{a+b}{ab}$$

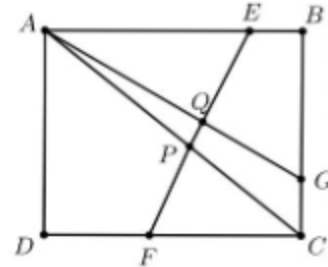
$$\frac{AF}{AB} = \frac{3}{8} = \frac{AF}{12}, \quad AF = \frac{3}{8} \times 12 = \frac{9}{2}$$



Math Exploration 2

1. Rectangle $ABCD$ has $AB = 5$ and $BC = 4$. Point E lies on $\overline{AB}$ so that $EB = 1$, point G lies on $\overline{BC}$ so that $CG = 1$ and point F lies on $\overline{CD}$ so that $DF = 2$. Segments $\overline{AG}$ and $\overline{AC}$ intersect $\overline{EF}$ at Q and P , respectively. (Adapted from 2016 AMC 10B Problem, Question #19)

(1) What is the value of $\frac{EP}{EF}$?



Math Exploration 2

1. Rectangle $ABCD$ has $AB = 5$ and $BC = 4$. Point E lies on $\overline{AB}$ so that $EB = 1$, point G lies on $\overline{BC}$ so that $CG = 1$ and point F lies on $\overline{CD}$ so that $DF = 2$. Segments $\overline{AG}$ and $\overline{AC}$ intersect $\overline{EF}$ at Q and P , respectively. (Adapted from 2016 AMC 10B Problem, Question #19)

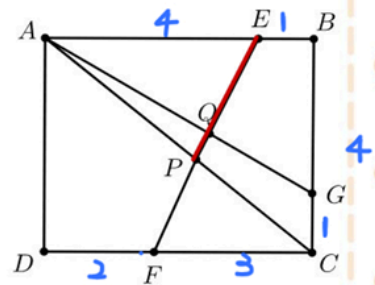
✓ (1) What is the value of $\frac{EP}{EF}$? $EF = EP + FP$, $\frac{EP}{FP} = ?$

$\triangle AEP \sim \triangle CFP$ "8" model

$$\frac{EP}{FP} = \frac{AE}{CF} = \frac{4}{3}$$

$$EF = 4 + 3 = 7$$

$$\frac{EP}{EF} = \frac{4}{7}$$



Math Exploration 3

3. Rectangle $ABCD$ has $AB = 8$ and $BC = 6$. Point M is the midpoint of diagonal $\overline{AC}$, and E is on $\overline{AB}$ with $\overline{ME} \perp \overline{AC}$. What is the area of $\triangle AME$? (2009 AMC 10B Problem, Question #18)

A. $\frac{65}{8}$

B. $\frac{25}{3}$

C. 9

D. $\frac{75}{8}$

E. $\frac{85}{8}$

Math Exploration 3

3. Rectangle $ABCD$ has $AB = 8$ and $BC = 6$. Point M is the midpoint of diagonal $\overline{AC}$, and E is on $\overline{AB}$ with $\overline{ME} \perp \overline{AC}$. What is the area of $\triangle AME$? (2009 AMC 10B Problem, Question #18)

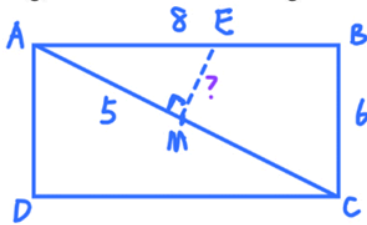
A. $\frac{65}{8}$

B. $\frac{25}{3}$

C. 9

D. $\frac{75}{8}$

E. $\frac{85}{8}$



$$\text{Area} = \frac{1}{2} \cdot 5 \cdot \frac{15}{4} = \frac{75}{8}$$

$$AM \cdot EM \cdot \frac{1}{2}$$

$$AB = 8, BC = 6, AC = 10, AM = 5$$

$$\triangle AME \sim \triangle ABC$$

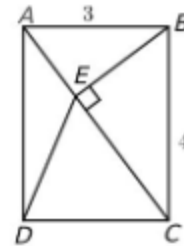
$$\frac{ME}{BC} = \frac{AM}{AB}$$

$$\frac{ME}{6} = \frac{5}{8}, ME = \frac{15}{4}$$

Math Exploration 4

2. Rectangle $ABCD$ has $AB = 3$ and $BC = 4$. Point E is the foot of the perpendicular from B to diagonal $\overline{AC}$. What is the area of $\triangle AED$? (2017 AMC 10B Problem, Question #15)

- A. 1 B. $\frac{42}{25}$ C. $\frac{28}{15}$ D. 2 E. $\frac{54}{25}$



Math Exploration 4

2. Rectangle $ABCD$ has $AB = 3$ and $BC = 4$. Point E is the foot of the perpendicular from B to diagonal $\overline{AC}$. What is the area of $\triangle AED$? (2017 AMC 10B Problem, Question #15)

- A. 1 B. $\frac{42}{25}$ C. $\frac{28}{15}$ D. 2 E. $\frac{54}{25}$

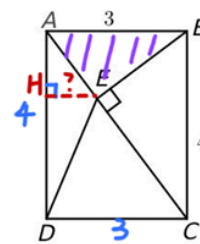
$EH \parallel DC$ $\triangle AHE \sim \triangle ADC$

$$\frac{EH}{DC} = \frac{AE}{AC} \quad \frac{EH}{3} = \frac{AE}{5} = \frac{9}{5}$$

$\triangle ABE \sim \triangle ABC$ $\frac{AE}{AB} = \frac{AB}{AC} = \frac{3}{5}$

$$AE = \frac{3}{5} \times AB = \frac{3}{5} \times 3 = \frac{9}{5}$$

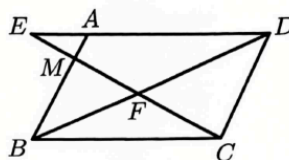
$$EH = \frac{9}{5} \times 3 = \frac{9}{5} \times \frac{1}{5} \times 3 = \frac{27}{25}$$



$$\text{Area} = \frac{1}{2} \cdot 4 \cdot \frac{27}{25} = \frac{54}{25}$$



- 1 In parallelogram $ABCD$, point E is on the extension of $\overline{DA}$. Then ____.



- A. $\frac{AE}{AD} = \frac{EM}{MC}$ B. $\frac{AE}{AD} = \frac{AM}{CD}$ C. $\frac{BM}{CD} = \frac{BF}{BD}$ D. $\frac{ED}{BC} = \frac{AD}{BM}$

A

Solution

A: since $ABCD$ is a parallelogram,

$\overline{AB} \parallel \overline{CD}$, $\overline{AD} \parallel \overline{BC}$,

so $\triangle AEM \sim \triangle DEC$.

Therefore $\frac{AE}{AD} = \frac{EM}{MC}$,

so A is correct;

B: since $\overline{AM} \parallel \overline{CD}$,

$\frac{AE}{ED} = \frac{AM}{CD}$.

So B is incorrect;

C: since $\overline{BM} \parallel \overline{CD}$,

$\triangle BMF \sim \triangle DCF$.

So $\frac{BM}{CD} = \frac{BF}{FD}$.

Therefore C is incorrect;

D: since $\overline{ED} \parallel \overline{BC}$,

$\triangle EFD \sim \triangle CFB$,

so $\frac{ED}{BC} = \frac{DF}{BF}$.

Since $\overline{AB} \parallel \overline{CD}$,

$\triangle BFM \sim \triangle DFC$,

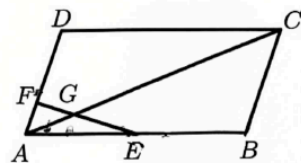
so $\frac{CD}{BM} = \frac{DF}{BF}$.

Therefore $\frac{ED}{BC} = \frac{CD}{BM}$,

so D is incorrect.

So the correct answer is A.

- 2 As shown in the figure, in parallelogram $ABCD$, E is the midpoint of $\overline{AB}$, and F is a point on $\overline{AD}$. $\overline{EF}$ intersects with $\overline{AC}$ at point G . If $AF = 2\text{cm}$, $DF = 4\text{cm}$, and $AG = 3\text{cm}$, then the length of $\overline{AC}$ is _____.



A. 14cm

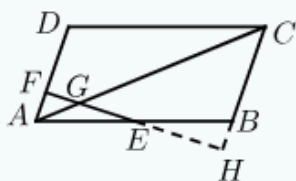
B. 15cm

C. 16cm

D. $\frac{46}{3}\text{cm}$

B

Solution



Since quadrilateral $ABCD$ is a parallelogram,

$BC = AD = 6\text{cm}$, and $\overline{BC} \parallel \overline{AD}$.

Therefore $m\angle BAF = m\angle EBH$, $m\angle AFE = m\angle BHE$.

Note that since $AE = BE$,

$\triangle AFE \cong \triangle BHE$,

so $BH = AF = 2\text{cm}$.

Since $\overline{BC} \parallel \overline{AD}$,

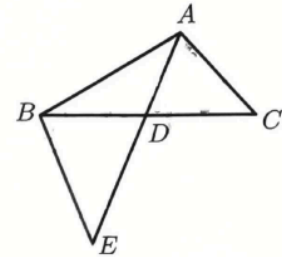
we have $\frac{AG}{CG} = \frac{AF}{CH}$, which means that $\frac{3}{CG} = \frac{2}{8}$.

Then $CG = 12$,

so $AC = AG + CG = 15(\text{cm})$.

Therefore, the correct answer is B.

- 9 As shown in the figure, in $\triangle ABC$, $\overline{AE}$ intersects with $\overline{BC}$ at point D . Draw $\overline{BE}$. If $m\angle C = m\angle E$, $AD : DE = 3 : 5$, $AE = 8$, and $BD = 4$, then the length of $\overline{DC}$ is _____.



A. $\frac{15}{4}$

B. $\frac{12}{5}$

C. $\frac{20}{3}$

D. $\frac{17}{4}$

Answer

A

Solution

Since $m\angle C = m\angle E$, and $m\angle ADC = m\angle BDE$,
 $\triangle ADC \sim \triangle BDE$,

so $\frac{DC}{DE} = \frac{AD}{BD}$.

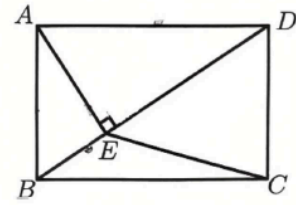
Also since that $AD : DE = 3 : 5$, and $AE = 8$,
 $AD = 3$, and $DE = 5$.

Since $BD = 4$,

we have $\frac{DC}{5} = \frac{3}{4}$,

so $DC = \frac{15}{4}$.

- 10 In rectangle $ABCD$, $\overline{AE} \perp \overline{BD}$ at point E . $AD = 3$, $BD = \sqrt{13}$, and draw $\overline{CE}$. Then the area of $\triangle DEC$ is _____.



Answer

$$\frac{27}{13} \text{ or } 2\frac{1}{13}$$

Solution

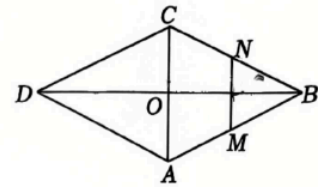
$$AD^2 = DE \times BD,$$

$$DE = \frac{AD^2}{BD} = \frac{9\sqrt{13}}{13}.$$

$$AE = \sqrt{3^2 - \left(\frac{9\sqrt{13}}{13}\right)^2} = \frac{6\sqrt{13}}{13},$$

$$\text{so } S_{\triangle DEC} = \frac{1}{2} \cdot DE \cdot AE = \frac{1}{2} \times \frac{9\sqrt{13}}{13} \times \frac{6\sqrt{13}}{13} = \frac{27}{13}.$$

- 3 As shown in the figure, O is the diagonal of rhombus $ABCD$ with AC as its intersection point. Points M and N are the midpoints of sides AB and BC , respectively, connecting MN . If $MN = 4$ and $BD = 12$, then the perimeter of rhombus $ABCD$ is _____.



Answer

$$8\sqrt{13}$$

Solution

From the problem, we know MN is the median of $\triangle ABC$.

$$\therefore AC = 2MN = 2 \times 4 = 8.$$

$\therefore$ Quadrilateral $ABCD$ is a rhombus,

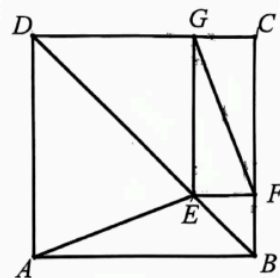
$$\therefore OA = \frac{1}{2}AC = \frac{1}{2} \times 8 = 4, OB = \frac{1}{2}BD = \frac{1}{2} \times 12 = 6, AC \perp BD.$$

In $Rt \triangle OAB$, by the Pythagorean theorem, we have $AB = \sqrt{OA^2 + OB^2} = \sqrt{4^2 + 6^2} = 2\sqrt{13}$.

$\therefore$ the perimeter of rhombus $ABCD$ is $4 \times 2\sqrt{13} = 8\sqrt{13}$.

Thus, the answer is: $8\sqrt{13}$.

- 8 As shown in the figure, point E is a point on the diagonal BD of square $ABCD$, $EF \perp BC$, $EG \perp CD$ at F , G , $GF = 3$ respectively, then $AE =$ _____ .



Answer

3

Solution

Connect CE ,

$\because BD$ is the diagonal of a square,

$\therefore \angle BCD = 90^\circ$, $\angle ABE = \angle CBE = 45^\circ$, $AB = BC$,

In $\triangle ABE$ and $\triangle CBE$,

$$\begin{cases} AB = CB \\ \angle ABE = \angle CBE, \\ BE = BE \end{cases}$$

$\therefore \triangle ABE \cong \triangle CBE (SAS)$,

$\therefore AE = CE$,

$\because EF \perp BC$, $EG \perp CD$,

$\therefore \angle EGC = \angle CFE = 90^\circ$,

$\therefore \angle EGC = \angle CFE = \angle BCD = 90^\circ$,

$\therefore$ Quadrilateral $EFCG$ is a rectangle,

$\therefore CE = FG = AE = 3$.

Math Exploration 1

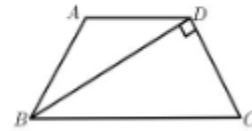
2. As shown in the figure, in the quadrilateral $ABCD$, $AD \parallel BC$, $AB = AD = DC = 1$, $BD \perp CD$, what is the area of the quadrilateral $ABCD$?

A. $\frac{\sqrt{3}}{3}$

B. $\frac{3\sqrt{3}}{2}$

C. $\frac{3\sqrt{3}}{4}$

D. $\sqrt{3}$



Math Exploration 1

2. As shown in the figure, in the quadrilateral $ABCD$, $AD \parallel BC$, $AB = AD = DC = 1$, $BD \perp CD$, what is the area of the quadrilateral $ABCD$?

A. $\frac{\sqrt{3}}{3}$

B. $\frac{3\sqrt{3}}{2}$

C. $\frac{3\sqrt{3}}{4}$

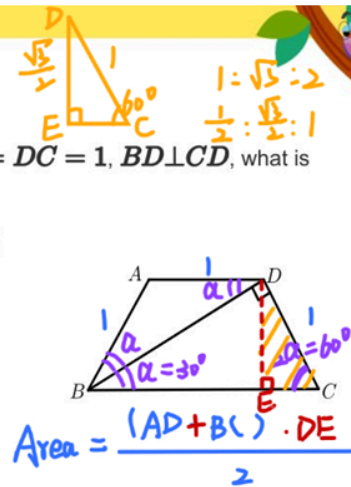
D. $\sqrt{3}$

iso $\triangle ABD$, $m\angle ABD = m\angle ADB = m\angle DBC = \alpha^\circ$
Since $AD \parallel BC$

$m\angle C = 2\alpha$, iso trapezoid

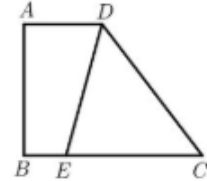
$\alpha + 2\alpha + 90 = 180$, $Rt\triangle BCD$ $30-60-90$, $DC=1$, $BC=2$
 $\alpha = 30$ $Rt\triangle CDE$ $30-60-90$, $DC=1$, $DE = \frac{\sqrt{3}}{2}$

Area = $\frac{(1+2) \cdot \frac{\sqrt{3}}{2}}{2} = \frac{3\sqrt{3}}{4}$



Math Exploration 1

3. As shown in the diagram, in the given trapezoid $ABCD$, $AD \parallel BC$, $\angle B = 90^\circ$, $CD = 15$, $BC = 16$, $AB = 12$. Point E is a point on side BC . Connect DE , and meet $DE = CE$. The area of trapezoid $ABCD$ is _____.



Math Exploration 1

3. As shown in the diagram, in the given trapezoid $ABCD$, $AD \parallel BC$, $\angle B = 90^\circ$, $CD = 15$, $BC = 16$, $AB = 12$. Point E is a point on side BC . Connect DE , and meet $DE = CE$. The area of trapezoid $ABCD$ is 138.

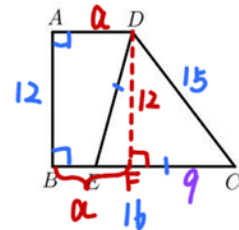
$DF \perp BC$,

$ABFD$ Rectangle, $DF = 12$, $BF = AD = a$

$Rt\triangle CDF$, $CD = 15$, $DF = 12$, $CF = 9$

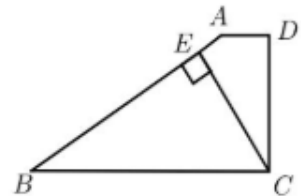
$BF = BC - CF = 16 - 9 = 7$, $BF = AD = 7$

$$\text{Area } ABCD = \frac{(16 + 7) \times 12}{2} = 138$$



Math Exploration 1

4. As shown in the figure, in the trapezoid $ABCD$, $AD \parallel BC$, $\angle ADC = 90^\circ$, $\angle B = 30^\circ$, $CE \perp AB$, with the perpendicular foot at point E . If $AD = 1$, $AB = 2\sqrt{3}$, find the length of CE .



Math Exploration 1

4. As shown in the figure, in the trapezoid $ABCD$, $AD \parallel BC$, $\angle ADC = 90^\circ$, $\angle B = 30^\circ$, $CE \perp AB$, with the perpendicular foot at point E . If $AD = 1$, $AB = 2\sqrt{3}$, find the length of CE .

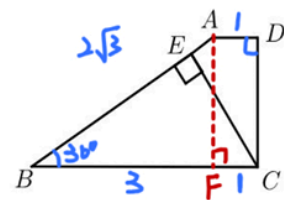
$AF \perp BC$

$30^\circ - 60^\circ - 90^\circ$ Rt $\triangle ABF$, $AB = 2\sqrt{3}$, $AF = \sqrt{3}$
 $BF = 3$

rectangle $AFCD$, $CF = AD = 1$

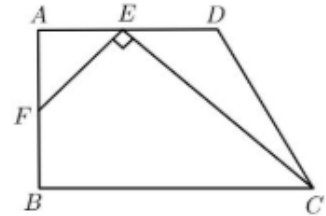
$BC = 4$

Rt $\triangle BCE$, $30^\circ - 60^\circ - 90^\circ$, $BC = 4$, $CE = 2$



Math Exploration 1

5. As shown in the figure, in trapezoid $ABCD$, $AD \parallel BC$, $AB \perp BC$, $AB = AD = 2$, $BC = 3$, point E is the midpoint of side AD . Through point E , draw $EF \perp EC$ intersecting AB at point F . Find the length of segment AF .



Math Exploration 1

5. As shown in the figure, in trapezoid $ABCD$, $AD \parallel BC$, $AB \perp BC$, $AB = AD = 2$, $BC = 3$, point E is the midpoint of side AD . Through point E , draw $EF \perp EC$ intersecting AB at point F . Find the length of segment AF .

$ABFE$ rectangle $EG = 2$ $BG = 1$

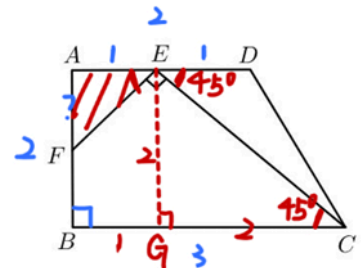
$$CG = BC - BG = 2$$

$\triangle CEG$ is iso $\text{Rt}\triangle$, $m\angle ECG = 45^\circ$

$AD \parallel BC$, $m\angle DEC = 45^\circ$

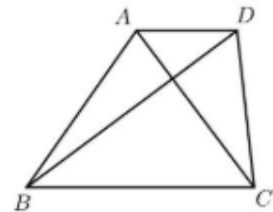
$$m\angle AEF = 180^\circ - 90^\circ - 45^\circ = 45^\circ$$

$\triangle AEF$ is iso $\text{Rt}\triangle$, $AF = 1$



Math Exploration 2

3. As shown in the figure, in trapezoid $ABCD$, $\overline{AD} \parallel \overline{BC}$, $AD = 2$, $BC = 8$, $AC = 6$, and $BD = 8$.
The area of trapezoid $ABCD$ is _____.



Math Exploration 2

3. As shown in the figure, in trapezoid $ABCD$, $\overline{AD} \parallel \overline{BC}$, $AD = 2$, $BC = 8$, $AC = 6$, and $BD = 8$.
The area of trapezoid $ABCD$ is 24.

$DE \perp BC$

$ACFD$ is parallelogram, $\begin{cases} AC \parallel DF \\ AD \parallel BC \Rightarrow AD \parallel CF \end{cases}$

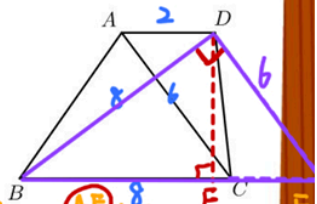
$$DF = AC = 6, CF = AD = 2$$

$$BF = BC + CF = 8 + 2 = 10$$

$$\triangle BDF, BD = 8, DF = 6, BF = 10,$$

$$BD^2 + DF^2 = BF^2, \text{ Right Triangle}$$

$$\begin{aligned} \text{Area } \triangle BDF &= \frac{8 \times 6}{2} = 24 \\ \text{Area } ABCD &= \frac{(2+8) \cdot 4.8}{2} = 24 \end{aligned}$$

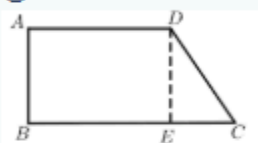


- 4 In the right trapezoid $ABCD$, $AD \parallel BC$, $\angle A = 90^\circ$, $AB = \frac{5\sqrt{3}}{2}$, $CD = 5$, then $\angle C =$ _____ .

60° or 120°

Solution

①As shown in the diagram 1,



Through point D , draw $DE \perp BC$ passing through point E , then

$$\angle DEC = \angle DEB = 90^\circ,$$

$$\because AD \parallel BC, \angle A = 90^\circ,$$

$$\therefore \angle B = 90^\circ,$$

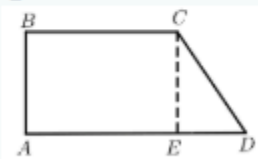
$\therefore$ the quadrilateral $ABED$ is a rectangle,

$$\therefore \angle ADE = 90^\circ, AB = DE = \frac{5\sqrt{3}}{2},$$

$$\because CD = 5,$$

$$\therefore \angle C = 60^\circ.$$

②As shown in the diagram 2,



At this point, it can be known from (i) that $\angle D = 60^\circ$,

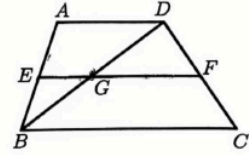
$$\because BC \parallel AD,$$

$$\therefore \angle BCD + \angle D = 180^\circ,$$

$$\angle BCD = 180^\circ - 60^\circ = 120^\circ.$$

That is, the degree of $\angle C$ is 60° or 120°

- 5 As shown in the figure, the median of a trapezoid has a length of 20cm. It is divided in two parts by the diagonal, such that the difference between the lengths of two parts is 5cm. Then the length of the longer base is _____ cm.



Answer

25

Solution

Since $GF - GE = 5\text{cm}$,

$$\frac{1}{2}BC - \frac{1}{2}AD = 5\text{cm},$$

so $BC - AD = 10\text{cm}$.

Since $EF = 20\text{cm}$,

$$AD + BC = 40\text{cm}.$$

therefore $BC = 25\text{cm}$.

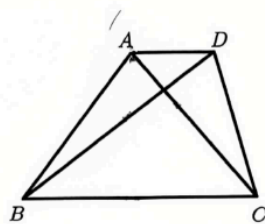
- 6 As shown in the figure, in trapezoid $ABCD$, $\overline{AD} \parallel \overline{BC}$, $AD = 2$, $BC = 3$, $AC = 3$, and $BD = 4$. The area of $ABCD$ is _____.

A. 20

B. 12

C. 24

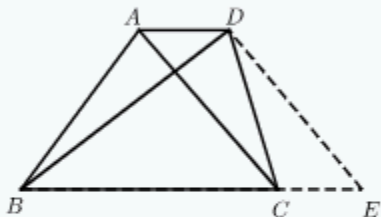
D. 6



D

Solution

Draw $\overline{DE} \parallel \overline{AC}$ through D , intersecting with the extension of $\overline{BC}$ at point E .



Since $\overline{AD} \parallel \overline{BC}$,

$ACED$ is a parallelogram,

so $DE = AC = 3$, $CE = AD = 2$.

Therefore $BE = BC + CE = 5$.

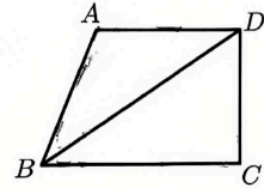
Also since $BD = 4$,

we have $m\angle BDE = 90^\circ$.

$$\text{So } S_{ABCD} = S_{\triangle BDE} = \frac{1}{2} BD \cdot DE = \frac{1}{2} \times 3 \times 4 = 6.$$

So the correct answer is D.

- 8 As shown in the figure, in the right trapezoid $ABCD$, $AD \parallel BC$, $\angle C = 90^\circ$, $\angle A = 120^\circ$, $AD = 2$, BD bisects $\angle ABC$, then the perimeter of trapezoid $ABCD$ is _____.



Answer

$$7 + \sqrt{3}$$

Solution

Pass through point A and draw $AE \perp BD$ at point E ,

$$\because AD \parallel BC, \angle A = 120^\circ,$$

$$\therefore \angle ABC = 60^\circ, \angle ADB = \angle DBC,$$

$$\because BD \text{ bisects } \angle ABC,$$

$$\therefore \angle ABD = \angle DBC = 30^\circ,$$

$$\therefore \angle ABE = \angle ADE = 30^\circ,$$

$$\therefore AB = AD,$$

$$\therefore AE = \frac{1}{2}AD = 1,$$

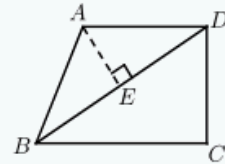
$$\therefore DE = \sqrt{3}, \text{ then } BD = 2\sqrt{3},$$

$$\because \angle C = 90^\circ, \angle DBC = 30^\circ,$$

$$\therefore DC = \frac{1}{2}BD = \sqrt{3},$$

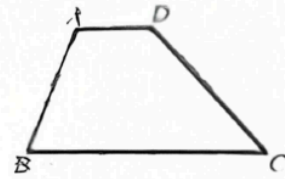
$$\therefore BC = \sqrt{BD^2 - CD^2} = \sqrt{(2\sqrt{3})^2 - (\sqrt{3})^2} = 3,$$

$$\therefore \text{the perimeter of trapezoid } ABCD \text{ is: } AB + AD + CD + BC = 2 + 2 + \sqrt{3} + 3 = 7 + \sqrt{3}.$$





It is given that the lengths of the four sides of a trapezoid are 1, 2, 3, and 4. Calculate the area of this trapezoid.



Answer

$$\frac{10\sqrt{2}}{3}$$

- 9 It is given that the lengths of the four sides of a trapezoid are 1, 2, 3, and 4. Calculate the area of this trapezoid.

Make $\overline{DE} \parallel \overline{AB}$, $\overline{DF} \perp \overline{BC}$

then $DE = 2$, $BE = 1$, $CE = 3$

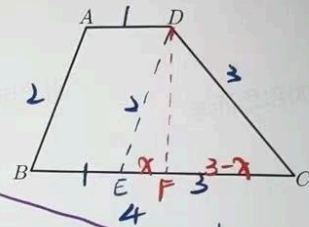
Assume $EF = x$, $CF = 3 - x$

In $\triangle DEF$, $2^2 - x^2 = DF^2$

In $\triangle CDF$, $3^2 - (3 - x)^2 = DF^2$

Answer $\frac{10\sqrt{2}}{3}$

Then, $2^2 - x^2 = 3^2 - (3 - x)^2$
 $4 - x^2 = 9 - 9 + 6x - x^2$
 $4 - x^2 = 6x - x^2$
 $4 = 6x$
 $x = \frac{2}{3}$



when $x = \frac{2}{3}$

$$4 - \frac{4}{9} = DF^2$$

$$DF^2 = \frac{32}{9}$$

$$DF = \frac{4\sqrt{2}}{3}$$

- 10 In trapezoid $ABCD$, $AB \parallel CD$, $DC : AB = 1 : 2$, the points E and F are respectively the midpoints of the two legs BC and AD . Then $EF : AB$ is _____.

A. 1 : 4

B. 1 : 3

C. 1 : 2

D. 3 : 4

$$\begin{aligned} \text{Area } ABLD &= \frac{(1+4)}{2} \times \frac{4\sqrt{2}}{3} \\ &= \frac{10\sqrt{2}}{3} \end{aligned}$$